Comparison of Salient Pole Type Rotor and Smooth Cylindrical Type Rotor

Sl.No	Salient Pole Rotor	Cylindrical Rotor
1	The diameter of rotor is large	The diameter of rotor is small
2	Poles are projecting outside	Poles are not projecting outside
3	Length of rotor is short	Length of rotor is long
4	Damper winding is required	No damper winding is required
5	Runs at slow speed. (100 rpm to 1500 rpm)	Runs at very high speed. (1500 rpm to 3000 rpm)
6	Suitable for hydro generators	Suitable for turbo alternators run by steam turbines

The field windings are accommodated in these slots. Generally copper strips are used for the field winding, Radial ducts are provided for ventilation purposes.

8.3.3 Parts of A.C Motors

1. Frame: It is made up of alloy cast iron.
2. Stator and Rotor core: It is built from high quality, low loss silicon steel laminations.
3. Stator and Rotor windings: They have moisture proof insulation embodying mica and high quality varnishes. They are carefully spaced for most effective air circulation.
4. Air gap: The stator is machined carefully to ensure uniformity of air gap.
5. Shafts and Bearings: The motor shaft is fixed with in the rotor and rotates with it. The rotor shaft is held in place by bearings at either end of the motor casing. Ball and roller bearings are used.
6. Fans: Light aluminum fans are used for adequate circulation of cooling air.

8.4 SINGLE PHASE AC MOTOR

Single phase and three phase system is widely used for domestic and commercial purpose. As the single-phase system is more economical and the power requirement in most of the houses, shops, offices are small, which can be easily met by three phase system.

Normally single phase motors are used for domestic purposes. Generally below 1 HP (1HP = 746 watts) motors are used for domestic purpose and they are called as fractional Horse Power motors (FHP)

Advantages

i) Simple in construction
ii) Cheap in cost
iii) Very reliable
iv) Easy to repair and maintenance

Applications

Due to all the above advantages, the single phase motor find its applications in

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